Chatbot (Rule-Based / Intent-Based):14/10/25(Tuesday)

Short summary :

This project involved the development of a Rule-Based/Keyword-Driven Intent Chatbot using Python and the Natural Language Toolkit (NLTK) library. The chatbot's primary function is to interpret the user's conversational intent (e.g., get_hours, bot_name) by analyzing keywords present in their input query.

The core solution relies on Natural Language Processing (NLP) techniques like Tokenization and Stemming to clean and standardize user text. Intent is determined by a simple Scoring Algorithm that counts matches between the processed user input and predefined intent keywords. This approach provides a robust, predictable, and maintainable foundational conversational interface.

Report:

1. Project Goal and Approach

The objective was to implement a functional, basic chatbot capable of answering a fixed set of Frequently Asked Questions (FAQs) in a conversational manner.

Aspect	Description
Project Type	Conversational AI (Foundational)
Core Approach	Intent-Based Classification using Keyword Matching
Technology	Python, NLTK (for NLP preprocessing)
Functionality	Determine user intent, provide random but relevant responses, and implement graceful fallbacks.

2. Implementation Steps (Aligned with Checklist)

The project followed a structured development pipeline:

1. Intent Definition: Defined specific user goals (e.g., greeting, get_hours) and a dictionary of associated responses and keywords.
2. Preprocessing Pipeline (NLP):

- Normalization: Convert input to lowercase and remove punctuation.
- Tokenization: Break sentences into individual words.
- Stemming (NLTK): Reduce words to their root form (e.g., 'opening' → 'open').

3. Intent Matching: Implemented a Scoring Algorithm to link the processed user tokens to the most likely intent.
4. Response Generation: For the classified intent, a response was selected randomly from a predefined list to simulate human variety.
5. Testing and Documentation: Rigorous testing with varied user queries and inclusion of a fallback response for unrecognized input.

3. Core Scoring Algorithm (Mathematical Principle)

The mechanism for classifying user input into a specific intent is based purely on counting keyword occurrences.

Variable	Definition
U	The set of stemmed tokens from the user's input.
Ki	The set of stemmed keywords defined for Intenti.
Ti	The required score threshold for Intenti.
Scorei	The calculated matching score for Intenti.
Match	The final predicted intent.

A. Score Calculation Formula

The match score for an intent is simply the count of common elements between the user's tokens (U) and the intent's keywords (Ki):

$Score_i = |U \cap K_i|$

B. Intent Classification Logic

The chatbot selects the intent that maximizes the score, provided that score meets a minimum required threshold (Ti):

$Match = \arg\max_i (Score_i)$

Subject to: $Score_i \geq T_i$

If no intent meets its respective threshold, the chatbot defaults to the Fallback Intent.

4. Key Takeaways and Limitations

Category	Finding/Limitation
Successes	Achieved accurate classification for all predefined queries; successfully used NLTK for preprocessing; maintained a clear, modifiable, rule-set.
Core Limitation	Lack of Context (Stateless): The bot cannot remember the history of the conversation, treating every query independently.
Scalability	Low: Adding new intents requires manually defining keywords and adjusting thresholds, which is time-consuming.
Robustness	Weak against Typos/Misspellings: Errors or new phrasing not covered by stemming can lead to a fallback.

This project serves as a solid foundation for future development using more advanced techniques like Machine Learning classifiers (e.g., Naive Bayes or Deep Learning) to improve the intent recognition process.